

Unit 1 - Number Sense and Operations With Positive Numbers

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The following B.E.S.T. Standards will be covered in this unit:

MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.

- *Clarification 1: Within this benchmark, expectations include finding greatest common factor within 1,000 and least common multiple with factors to 25.*
- *Clarification 2: Instruction includes finding the greatest common factor of the numerator and denominator of a fraction to simplify the fraction.*

MA.NSO.3.2 Rewrite the sum of two composite whole numbers that have a common factor, as a common factor multiplied by the sum of two whole numbers.

- *Clarification 1: Instruction includes using the distributive property to generate equivalent expressions.*

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

- *Clarification 1: Multi-digit decimals are limited to no more than 5 total digits.*

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.

- *Clarification 1: Instruction focused on making connections between visual models, the relationship between multiplication and division, reciprocals and algorithms*

Unit 1, Lesson 1: Finding the Greatest Common Factor



Benchmark

MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.



Learning Targets

- ☐ I can find the greatest common factor (GCF) in mathematical and real-world contexts.
- ☐ I can rewrite fractions in simplest form using the greatest common factor (GCF) of the numerator and denominator.



1.1.1 Warm-Up: Where Do Math Symbols Come From?

Math is full of symbols that include lines, dots, arrows, English letters, Greek letters, and more. It can look and sometimes feel like its own language, and in many ways it is!



Robert Recorde invented the equals sign, $=$, because he was tired of writing “is equal to” over and over again.

1. What is a word or phrase that you use a symbol to represent instead of writing it out each time?

2. Why did early mathematicians decide to use symbols to express operations, phrases, and mathematical ideas?

3. Why are math symbols still helpful to us today?





1.1.2 Guided Instruction: Greatest Common Factor

1. Marcus and Tonya are making care packages for soldiers overseas. Inside of each package, they will include popular snacks and handwritten letters. They have 16 snacks and 12 letters to put in the care packages. They want each care package to be equally sized, containing both snacks and letters, with nothing left over.

- a. Complete the tables below to show the possible combinations when making each care package.

Distributing Snacks Equally	
Number of Care Packages	Number of Snacks per Package
1	16
2	8
4	
	1

Distributing Letters Equally	
Number of Care Packages	Number of Letters per Package
1	12
2	6
3	
	3
	2
	1

- b. What is the greatest number of care packages that can be created using **only** snacks?

- c. What is the greatest number of care packages that can be created using **only** letters?

- d. What is the greatest number of care packages that can be created using **both** snacks and letters?



A **factor** is a number that divides into another number exactly and without leaving a remainder.

For example, the **factors** of 20 are 1, 2, 4, 5, 10, and 20.

The numbers (or expressions) that are being multiplied are called the factors. When numbers (or expressions) are multiplied, the answer is called a product.



factor $\times$ **factor** = **product**

- e. How were the factors of 12 and 16 used in the previous problem?
- The factors of 16 were used to determine the whole number of ☐ letters ☐ snacks that could be included, depending on the number of care packages.
 - The factors of 12 were used to determine the whole number of ☐ letters ☐ snacks that could be included, depending on the number of care packages.



The **greatest common factor**, or **GCF**, is the largest common factor of two or more numbers.

An example of how to find the **greatest common factor (GCF)** is shown.

The factors of 15 are 1, 3, 5, and 15.

The factors of 20 are 1, 2, 4, 5, 10, and 20.

The GCF of 15 and 20 is 5.

- f. Complete the statement to explain what the GCF of 16 and 12 is from the previous problem.

The GCF of 16 and 12 is _____, which represents the greatest number of care packages that can be created using

- ☐ only snacks.
☐ only letters.
☐ both snacks and letters.

2. Find the greatest common factor of 6 and 18.

	Factors	GCF:
6		
18		



1.1.3 Try It: Greatest Common Factor

- Find the greatest common factor of each pair of numbers.
 - 15 and 200
 - 123 and 41
 - 54 and 36
 - 282 and 108
- Penelope is organizing the teacher-student relay race. The teams must be divided evenly, with no participants left out. If there are 56 students registered and 21 teachers, what is the greatest number of teacher-student teams that will be able to participate in the relay race?



1.1.4 Guided Instruction: Writing Equivalent Fractions Using GCF

The GCF can also be used to rewrite fractions. When using the GCF to write an equivalent fraction, the result is often said to be in “simplest” form.



To **simplify a fraction**, generate an equivalent fraction by dividing the numerator and denominator by the **GCF** of the numerator and denominator.

- In the fraction $\frac{25}{90}$, the numerator is 25 and the denominator is 90.
 - What is the greatest common factor of 25 and 90?
 - Divide the numerator and denominator by the GCF. What is the resulting fraction?

- c. Complete the statement to describe the relationship between the fractions.

After dividing the numerator and denominator of $\frac{25}{90}$

by their GCF, the resulting fraction, ,

- ☐ is
☐ is not

equivalent to $\frac{25}{90}$, rewritten in simplest form.



1.1.5 Try It: Writing Equivalent Fractions using GCF

1. Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

a. $\frac{8}{28}$

b. $\frac{45}{90}$

c. $\frac{16}{34}$

2. Mikayla states that $\frac{900}{1000}$ is not the same value as $\frac{9}{10}$, because they are fractions of different sizes. Is she correct? Provide an explanation to support your answer.



1.1.6 Your Turn!

- Find the greatest common factor (GCF) of each pair of numbers.
 - 18 and 24
 - 45 and 60
 - 665 and 30
- Rewrite each fraction in simplest form using the GCF of the numerator and denominator.
 - $\frac{29}{87}$
 - $\frac{10}{15}$
 - $\frac{175}{210}$
- Mrs. Chaunuan is making gift bags filled with pencils and stickers for class prizes. Mrs. Chaunuan has 24 pencils and 36 stickers to use. Each bag will have the same number of each item, with no items left over. What is the greatest number of gift bags she can make?



1.1.7 Wrap-Up: Ticket Out the Door

- Rewrite $\frac{52}{104}$ in simplest form.
- Diego is preparing brownies and cookies for a bake sale. He has 48 brownies and 64 cookies. Diego wants to make bake sale bags. Each bag will have the same amount of each item, with no items left over. What is the greatest number of bake sale bags he can make?

Unit 1, Lesson 2: Finding the Least Common Multiple



Benchmark

MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.



Learning Target

- ☐ I can find the least common multiple (LCM) in mathematical and real-world contexts.

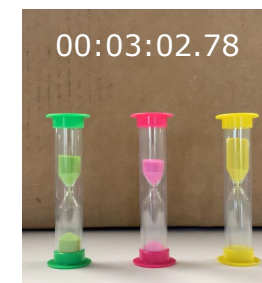
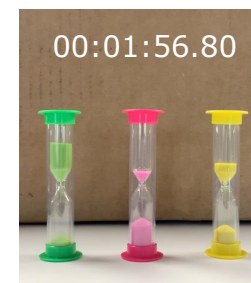
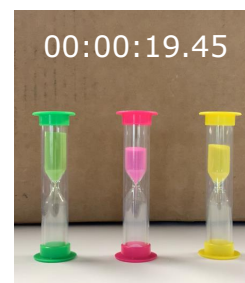


1.2.1 Warm-Up: Hourglass Timing

The video shows three different hourglass sand timers. The timers are all started at the same time. As one of the times runs out, it is flipped over again to restart.



Several screenshots taken from the video are shown.



- Write one question you have as you watch the video or review the images.
- Compare questions with your partner.
- Record your partner's question in the table and have them initial that you recorded it correctly. If you had the same question, work with your partner to generate a different question to ask about the video. *You are the only person who should write in the first column of the table.*

My Partner's Question From the Video	Initials



1.2.2 Guided Instruction: Least Common Multiple

1. Three timers were used in the video, a 2-, 3-, and 5-minute timer.
 - a. On each number line model, plot a point indicating the time at which each hourglass would need to be flipped over to restart the timer during a 30 minute period.

Time	Points Where the Timer Needs to be Restarted
2-minute timer	
3-minute timer	
5-minute timer	

- b. What do you notice about the times plotted on the number line for each timer?

Multiples of positive whole numbers are the product of that number multiplied by a positive whole number.

For example, the **multiples** of 2 are 2, 4, 6, 8,



The **ellipses** (...) at the end of the list of multiples means that the list continues beyond the last number listed. There are an **infinite** number of multiples for any given whole number.

2. Use the number line models for each of the timers and the definition of multiples to answer the questions.

- a. What are all the common multiples among the three timers?
 - b. What do these common multiples represent in this scenario?
 - c. After how many minutes will the 2- and 5-minute timers run out of sand at the same time?
 - d. After how many minutes will the 3- and 5-minute timers run out of sand at the same time?



The **Least Common Multiple**, or **LCM**, is the lowest number that is a multiple of two or more given numbers.

3. Use the definition to complete the statement.

The least common multiple (LCM) of 2, 3, and 5 is _____.

In the context of the hourglass timers, this value

represents _____

_____.

4. Find the LCM of 8 and 6.



1.2.3 Try It: Least Common Multiple

1. Find the LCM of each pair of numbers.

a. 10 and 4 b. 5 and 13 c. 7 and 21

2. Every 4 weeks, Fort Caroline Middle has an early release day. Every 5 weeks, they host an assembly with a guest speaker for all students. After how many weeks will Fort Caroline Middle have an assembly and early release in the same week?

3. A radio station is running two ongoing promotions. For the first promotion, every 30th caller wins a pair of movie tickets. For the second promotion, every 100th caller wins an annual pass to a theme park. Which caller number would be eligible for both prizes?

4. Vincenzo and Marie both joined a local gym. Vincenzo plans to go to the gym every 2 days at 7:00 am when they open. Marie plans to go every 3 days at 7:00 am. If they both joined the gym on the same day, after how many days will they arrive at the gym on the same day?



1.2.4 Collaboration: Least Common Multiple

1. Place a digit in each blue box to make a true statement. Use only digits 0 to 9, at most one time each.

Factor A:

Factor B:

LCM:

2. What were some of the challenges you faced with this problem?
3. What methods did you use to solve this problem?



1.2.5 Your Turn!

1. Find the LCM of each pair of numbers.
 - a. 15 and 20
 - b. 18 and 24
 - c. 22 and 11

2. Jasmine and Teia are running a race. Jasmine runs at a rate of 4 minutes per lap and Teia runs at a rate of 6 minutes per lap. If they both start running at the same time, after how many minutes will they both be at the starting line at the same time again?

3. Lin's uncle is opening a bakery. On the bakery's grand opening day, he plans to give away prizes to the first 50 customers that enter the shop. Every 9th customer will get a free blueberry muffin. Every 12th customer will get a free slice of carrot cake. Which customer will get both free bakery items?



1.2.6 Wrap-Up: Reflection

1. Rate your level of mathematical confidence on the concepts listed by circling the statement that best describes your understanding.

- **Greatest Common Factor (GCF)**

I don't know what GCF is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.
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- **Least Common Multiple (LCM)**

I don't know what LCM is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.
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Unit 1, Lesson 3: Rewriting Expressions Using the Distributive Property



Benchmark

MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.



Learning Targets

- ☐ I can use the distributive property to generate equivalent numeric expressions.
- ☐ I can rewrite the sum of two composite numbers with a common factor using the distributive property.



1.3.1 Warm-Up: Secret Number

Juanita has a secret number. Read her clues and answer the questions that follow.

Clue 1: My secret number is a factor of 60.

1. What is the smallest value Juanita's number could be?
2. What is the largest value Juanita's number could be?

Clue 2: My number is prime.

3. What values are possible for Juanita's number at this point?

Clue 3: 15 is a multiple of my secret number.

Clue 4: My secret number is a factor of 20.

4. What is Juanita's secret number?
5. Could you have determined her secret number only using the first three clues? Explain.



1.3.2 Guided Instruction: The Distributive Property



The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property "distributes" a factor to all terms within a set of parentheses.

$$a \times (b + c) = (a \times b) + (a \times c)$$

The product 7×23 can be rewritten using the distributive property by decomposing 23 into two parts and multiplying 7 by both parts.

$$\begin{aligned} 7 \times 23 &= 7(20 + 3) \\ 7(20 + 3) &= (7 \cdot 20) + (7 \cdot 3) \end{aligned}$$

1. What are two other ways that 7×23 could be rewritten using the distributive property?

$$7 \times 23 = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

$$7(23) = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

2. Which of the expressions is the easiest to evaluate using mental math?

The distributive property can also be used to rewrite the sum of two numbers that share a common factor.

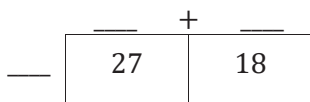
3. Elsee rewrote the sum $27 + 18$ as $3(9 + 6)$. Are these two expressions equivalent?

a. $27 + 18$ and $3(9 + 6)$ ☐ are
☐ are not equivalent expressions because when evaluated they have

- ☐ the same value.
☐ different values.

b. What common factor of 27 and 18 did Elsee use to rewrite the expression?

c. Rewrite the expression $27 + 18$ using a different common factor of 27 and 18. Use the area model to support your thinking.



$$27 + 18 = \underline{\hspace{1cm}}(\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$



1.3.3 Guided Practice: Rewriting Expressions

1. Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
98×5	$5 (90 + 8)$	5 90 450 + 8 40	
6×22			

2. Complete the table by rewriting each addition expression using the distributive property, modeling with area, and finding the sum.

Addition Expression	Distributive Property	Area Model	Sum
$600 + 140$		20 30 600 + 7 140	
$56 + 32$		— — + —	





1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property

Sums of **composite whole numbers** can be rewritten using the distributive property and a common factor of the numbers.



A **composite number** is a whole number greater than 1 that has at least one whole-number factor other than one and itself.

1. What are the common factors of 16 and 24?

Factors of 16	
Factors of 24	
Common Factors of 16 and 24	

2. Isaiah rewrote the sum $16 + 24$ using the distributive property. Each step in his thinking is shown. Complete the descriptions of Isaiah's process by examining his thinking and work. Use the words in the bank below to complete each description.

distributive equivalent factor multiplication

Isaiah's Thinking (💡) and Work (✍️)	Description
 "I know 2 is a common factor of both 16 and 24."	Isaiah identified a _____ both numbers have in common.
 $16 + 24 = 2(8) + 2(12)$	He rewrote the expression using _____ with the common factor.
 $2(8) + 2(12) = 2(8 + 12)$	He rewrote the expression using the _____ property.
 "The sum $16 + 24$ can be rewritten as $2(8 + 12)$ using the distributive property."	Isaiah recognized that $16 + 24$ and $2(8 + 12)$ are _____ numeric expressions.

3. Rewrite the sum $16 + 24$ using the distributive property in two additional ways.

$$16 + 24 = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

$$16 + 24 = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$



4. Two students made errors rewriting the sum $54 + 36$ using the distributive property. Each student's expression is shown.

Rewrite $54 + 36$ using the distributive property.	
Naveah's Expression	$4(14 + 9)$
Gavin's Expression	$3(18 + 36)$

Work with your partner to answer the following questions.

- What error did Naveah make?
 - What error did Gavin make?
 - Generate two different equivalent expressions for the sum $54 + 36$ by rewriting it using the distributive property.
5. Rewrite an equivalent expression for the sum of $40 + 64$ using the distributive property with the **greatest common factor** (GCF).



1.3.5 Collaboration: Missing Values

1. Work with your partner to determine the missing values that make each equation true.

a. $40 + 16 = 4(10 + \underline{\hspace{1cm}})$ b. $15 + 80 = 5(\underline{\hspace{1cm}} + 16)$

c. $108 + 198 = \underline{\hspace{1cm}}(6 + 11)$ d. $28 + 63 = 7(\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$

e. $45 + 18 = \underline{\hspace{1cm}}(\underline{\hspace{1cm}} + 6)$ f. $135 + 27 = 9(\underline{\hspace{1cm}} + 3)$

2. Choose one of the equations from question 1 and explain how you determined the missing value(s).

To find the missing value in equation _____, I

_____.

3. Have your partner read your explanation and offer suggestions on how to improve it.



1.3.6 Your Turn!

1. Rewrite an equivalent expression for each sum using the distributive property with a common factor.
 - a. $51 + 21$
 - b. $124 + 20$
 - c. $72 + 24$
 - d. $39 + 65$
2. Rewrite an equivalent expression for the sum of $68 + 34$ using the distributive property with the **greatest common factor** (GCF).



1.3.7 Wrap-Up: Ticket Out the Door

1. Rewrite an equivalent expression for the sum of $75 + 30$ in two different ways using the distributive property and a common factor.
2. Complete the equation with the missing values that make the equation true.

$$49 + 35 = \underline{\hspace{1cm}} (7 + \underline{\hspace{1cm}})$$

Unit 1, Lesson 4: Multiplying Multi-Digit Numbers With Decimals to the Thousandths



Benchmark

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.



Learning Target

- ☐ I can fluently multiply multi-digit numbers with decimals to the thousandths.



1.4.1 Warm-Up: Decimal Products

- Place a digit in each blue box so that the product is as close to 50 as possible. Use only digits 1 to 9, at most one time each.

$$\boxed{}.\boxed{} \times \boxed{}.\boxed{} =$$



1.4.2 Refresher: Reasoning with Powers of Ten

Several expressions are shown.

$250 \cdot 0.01$	$250 \div 100$	$250 \cdot 0.001$	$250 \cdot \frac{1}{100}$	$250 \cdot 0.1$
$250 \div 1000$	$250 \cdot \frac{1}{10}$	$250 \cdot \frac{1}{1000}$	$250 \div 10$	

1. Work with your partner to complete the table by writing the expressions that match each value indicated.

Value	Equivalent Expressions		
25			
2.5			
0.25			

2. When working together through the problem, Cheryl and Miranda had a disagreement.

Cheryl said that $\frac{1}{100}$ is equivalent to 0.001 because both have two zeros.	Miranda said that $\frac{1}{100}$ is equivalent to 0.01 because of place value.
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- a. Which student do you agree with?
- b. Explain why you agree.

Recall that fractions like $\frac{1}{10}$, $\frac{1}{100}$, and $\frac{1}{1000}$ can be used to find the location of the decimal point in terms of place value when multiplying decimals.

For example, 24×0.02 can be thought of as $24 \times \frac{2}{100}$ or $24 \times 2 \times \frac{1}{100}$. This is equal to $\frac{48}{100}$ which is read, "48 hundredths" and written as 0.48 in decimal form with the 8 in the hundredths place.

$$24 \times 0.02 = 0.48$$

3. With your partner, complete the equations and statements below to show how similar thinking can be used to find the products of decimals.

a.

$$(0.7) \cdot (0.09) = (7 \times \frac{1}{10}) \cdot (9 \times \frac{1}{100}) = (7 \cdot 9) \cdot (\frac{1}{10} \cdot \frac{1}{100}) = \boxed{} \cdot \frac{1}{\boxed{}}$$

Written as a decimal, $(0.7) \cdot (0.09) = \underline{\hspace{2cm}}$.

b.

$$(1.5) \cdot (0.43) = (15 \times \frac{1}{10}) \cdot (43 \times \frac{1}{100}) = \boxed{} \cdot \frac{1}{\boxed{}}$$

Written as a decimal, $(1.5) \cdot (0.43) = \underline{\hspace{2cm}}$.

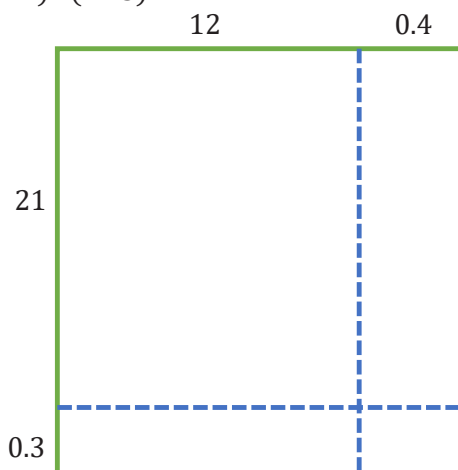




1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths

Different strategies can be used to find the product of decimals to the thousandths.

- Byanca prefers using an area model to find the product of decimals. Below is an area model that Byanca drew to represent $(12.4) \cdot (21.3)$.



- Find the region that represents $(0.4) \cdot (0.3)$. Shade that region in the diagram and label it with its area of 0.12.
- Label the other regions of the model with their respective areas.
- Find the value of $(12.4) \cdot (21.3)$ using the labeled area model. Show your work.

- Sergio prefers to use partial products to find the value of $(12.4) \cdot (21.3)$, while Alicia prefers using the standard algorithm. Their work is shown.

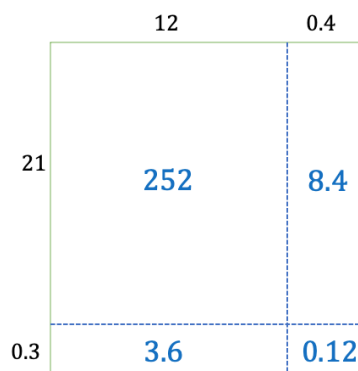
Sergio's Work Partial Products	Alicia's Work Standard Algorithm
$ \begin{array}{r} 12.4 \\ \times 21.3 \\ \hline 0.12 \\ 0.6 \\ 3 \\ 0.4 \\ 2 \\ 10 \\ 8 \\ 40 \\ + 200 \\ \hline 264.12 \end{array} $	$ \begin{array}{r} 12.4 \\ \times 21.3 \\ \hline 3.72 \\ 12.40 \\ + 248.00 \\ \hline 264.12 \end{array} $

Discuss each method with your partner while answering the following questions.

- In Sergio's work, which parts of the factors were multiplied to get each partial product indicated in the table below?

Partial Product	Parts of Factors Multiplied
0.12	___ $\times$ ___
0.6	___ $\times$ ___
3	___ $\times$ ___

- b. To which parts of the area model do these partial products relate?



- c. Where do the values of 3.72, 12.4, and 248 come from in Alicia's work that shows the standard algorithm?

- _____ is the product that results from multiplying by each part of 12.4.

- ☐ 0.3
- ☐ 1
- ☐ 20

- _____ is the product that results from multiplying by each part of 12.4.

- ☐ 0.3
- ☐ 1
- ☐ 20

- _____ is the product that results from multiplying by each part of 12.4.

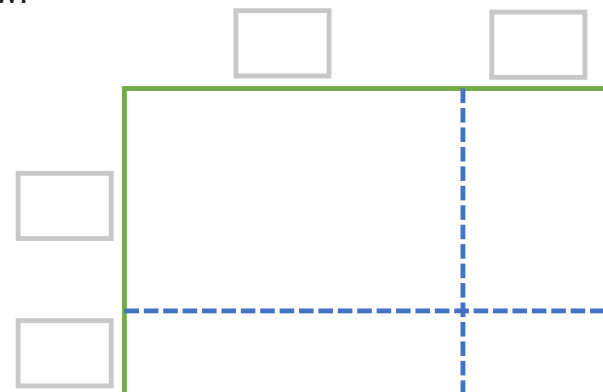
- ☐ 0.3
- ☐ 1
- ☐ 20

- d. Draw lines to indicate connections between the partial products found in each of Sergio and Alicia's methods.
- e. In each method, why are the numbers between the horizontal lines aligned vertically?



1.4.4 Guided Practice: Multiplying Decimals to the Thousandths

1. Find the product $(32.5)(6.02)$ using the area model below.



2. With your partner, determine who will be Partner A and Partner B.

- a. Find the product of 3.621×9.7 using the method indicated. Show your work in the space provided.

Partner A Partial Products	Partner B Standard Algorithm

- b. Compare your work with your partner to verify that you both found the same product. Discuss any errors and make corrections as needed. Once you have come to an agreement, write your partner's work in your workbook in the space provided.
- c. Discuss with your partner the strengths and weaknesses of using each method.
- d. Which method do you prefer for multiplying decimals to the thousandths?



1.4.5 Try It: Multiplying Decimals to the Thousandths

1. Find each product using your preferred method. Show your work.
 - a. $25.36 \cdot 14.8$
 - b. 19.008×3.2
 - c. $(36.3)(0.74)$



1.4.6 Your Turn!

1. Find each product.

a. 105.46×3.8

b. $8.7 \cdot 0.475$

c. $(7.7)(102.65)$

d. 1.561×12.8



1.4.7 Wrap-Up: Ticket Out the Door

1. Find each product.

a. $(6.36)(12.27)$

b. $8.12 \cdot 29.4$

Unit 1, Lesson 5: Dividing Multi-Digit Numbers With Decimals to the Thousandths



Benchmark

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.



Learning Target

- ☐ I can fluently divide multi-digit numbers with decimals to the thousandths.



1.5.1 Warm-Up: Quotients as Decimals

The quotients of several division expressions are shown.

$1 \div 2 = 0.5$	$1 \div 3 = 0.333\overline{3}$	$1 \div 4 = 0.25$
$1 \div 5 = 0.2$	$1 \div 6 = 0.666\overline{6}$	$1 \div 7 = 0.14285\overline{7}$
$1 \div 8 = 0.125$	$1 \div 9 = 0.111\overline{1}$	$1 \div 10 = 0.1$
$1 \div 11 = 0.0909\overline{09}$	$1 \div 12 = 0.083\overline{3}$	$1 \div 13 = 0.07692\overline{3}$

1. Write something you notice about the values of the quotients and something you still wonder.

I notice...	I wonder...





1.5.2 Refresher: Long Division

Jiang was dividing using long division and got stuck.

- Complete the problem to show Jiang the missing steps.

$$\begin{array}{r}
 15 \overline{) 1875} \\
 \underline{- 15} \\
 37 \\
 \underline{- 36} \\
 18 \\
 \underline{- 18} \\
 0
 \end{array}$$

- What advice about long division do you have that could help Jiang in the future?



1.5.3 Exploration: Dividing Decimals

Giovanny and Alina used different strategies to find the quotient of $48.78 \div 9$.

Giovanny's Strategy

$$\begin{array}{r}
 5.42 \\
 9 \overline{) 48.78} \\
 \underline{- 45} \\
 37 \\
 \underline{- 36} \\
 18 \\
 \underline{- 18} \\
 0
 \end{array}$$

Alina's Strategy

$$\begin{array}{r}
 5.42 \\
 900 \overline{) 4878} \\
 \underline{- 4500} \\
 3780 \\
 \underline{- 3600} \\
 1800 \\
 \underline{- 1800} \\
 0
 \end{array}$$

$$48.78 \div 9 = 5.42$$

Work with your partner to answer the questions about Giovanny and Alina's strategies.

- How are the two students' calculations alike and different?

Similarities	Differences

2. Look at Giovanni's calculations. Complete the statements with the appropriate place value to explain how you can tell that the 36 means "36 tenths" and the 18 means "18 hundredths" in his calculations.

- The 3 in 36 is in the _____ place and the 6 is in the _____ place, so 36 in this calculation means 36 tenths.
- The 1 in 18 is in the _____ place, so it has the same value as 10 hundredths. The 8 is in the _____ place. Together, they make 18 hundredths.

3. Look at Alina's calculations. Complete the statement to describe the values of 3600 and 1800 in her calculation.

- In Alina's calculations, the 3600 represents 3600 _____, and the 1800 represents 1800 _____.

4. Why is the value of $48.78 \div 9$ the same as the value of $4878 \div 900$?

5. Explain why $51.84 \div 6.4$ has the same value as $518.4 \div 64$.

6. Find the value of $51.84 \div 6.4$ using long division. Show your work.



1.5.4 Bring It Together: Dividing Decimals

1. Complete the summary of the exploration of dividing decimals using the words and phrases from the bank below.

does not change	multiply
power of 10	whole numbers

One way to find a quotient of two decimals is to first

_____ each decimal by a _____

so that both values are _____. If we

multiply both values by the same power of 10, this

_____ the value of the quotient.



1.5.5 Guided Practice: Dividing Decimals

1. Find each quotient using a strategy similar to Giovanni's or Alina's. Show your work.
 - a. $24.2 \div 1.1$

 - b. $17.028 \div 0.08$

2. Coby and Olivia each found the quotient of $10.92 \div 2.6$. The values they each used are listed.
 - Coby calculated $1092 \div 260$.
 - Olivia calculated $109.2 \div 26$.
 - a. Will the values they used result in the same quotient as $10.92 \div 2.6$? Explain.

 - b. Find the quotient of $10.92 \div 2.6$.



1.5.6 Your Turn!

1. Find each quotient. Show your work.
 - a. $43.396 \div 57.1$
 - b. $29.88 \div 0.36$

 - c. $262.5 \div 5.25$
 - d. $15.129 \div 12.3$



1.5.7 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate where your current learning is in relation to the learning target.

I can fluently divide multi-digit numbers with decimals to the thousandths.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



2. Briefly explain why you placed the X where you did.

Unit 1, Lesson 6: Multiplying Positive Fractions



Benchmark

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.



Learning Target

- ☐ I can fluently multiply positive fractions by positive fractions.



1.6.1 Warm-Up: Food Science

Dr. Charlotte Armah is a research scientist at the Institute of Food Research. She leads experiments involving human volunteers to learn whether or not eating particular foods can protect us from diseases such as cardiovascular disease and cancer. Her work allows her to combine her scientific and math expertise, interest in helping others, and enjoyment of interacting with different people.



1. What is one aspect of Charlotte's story that was surprising to you? Explain why this stood out to you.



1.6.2 Refresher: Multiplying Fractions



When **multiplying fractions**, multiply the numerators of the factors to find the numerator of the product, and multiply the denominators of the factors to find the denominator of the product.

$$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

1. With your partner, determine who will be partner A and who will be partner B.
 - a. Find the product $1\frac{2}{3} \times 3\frac{8}{9}$ using the method indicated.

Partner A: Area Model	Partner B: Rewrite the Mixed Number(s) as a Fraction Greater Than 1
	$1\frac{2}{3} \times 3\frac{8}{9}$

- b. Compare your work with your partner to verify that you both found the same product. Discuss any errors and make corrections as needed. Once you are in agreement, write your partner's work in your workbook in the space provided.



1.6.3 Guided Instruction: Multiplying Fractions

Recall from Lesson 1 that fractions can be rewritten in simplest form using the GCF of the numerator and denominator.

Find each product using the steps indicated.

1.

$$\frac{5}{8} \times 4\frac{2}{3}$$

Write the given multiplication expression.	
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	
Multiply to find the product of the fractions.	
Rewrite the product in simplest form using the GCF of the numerator and denominator.	

2.

$$3\frac{3}{5} \cdot 2\frac{1}{6}$$

Write the given multiplication expression.	
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	
Multiply to find the product of the fractions.	
Rewrite the product in simplest form using the GCF of the numerator and denominator.	



1.6.4 Try It: Multiplying Fractions

1. Find each product. Rewrite each product in simplest form. Show your work.

a. $\left(1\frac{1}{8}\right)\left(5\frac{5}{7}\right)$

b. $4\frac{4}{5} \times \frac{10}{13}$

c. $\frac{4}{11} \cdot \frac{1}{2}$

d. $2\frac{1}{3} \times 4\frac{4}{5}$

2. Complete the statement to describe which product was the most challenging for you to find.

The product of _____ was most challenging for me to find because _____.

3. What is a strategy that could help you on a similar problem in the future?



1.6.5 Your Turn!

1. Find each product. Simplify when possible.

a. $\frac{2}{3} \times \frac{5}{4}$

b. $2\frac{1}{3} \times \frac{3}{4}$

c. $\frac{8}{7} \cdot \frac{7}{10}$

d. $\frac{13}{2} \cdot \frac{7}{4}$

e. $7\frac{5}{6} \times 9$

f. $\left(\frac{24}{27}\right)\left(\frac{45}{32}\right)$

g. $\frac{3}{2} \cdot \frac{7}{10}$

h. $\left(2\frac{6}{25}\right)\left(\frac{1}{25}\right)$

i. $4\frac{1}{5} \times 1\frac{7}{9}$

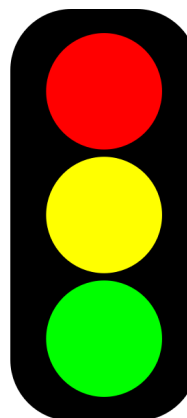


1.6.6 Wrap-Up: Reflection

In this unit, you have learned multiple concepts including those listed below.

- Finding the GCF and LCM of two or more numbers
- Rewriting sums using distributive property
- Multiplying and dividing decimals
- Multiplying fractions

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck about how to...

I'm starting to get the hang of...

I feel confident and ready to go with...

Unit 1, Lesson 7: Dividing Positive Fractions



Benchmark

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.



Learning Target

- ☐ I can fluently divide positive fractions by positive fractions.



1.7.1 Warm-Up: Error Analysis

Mateo tried to multiply $3\frac{5}{6} \cdot 4\frac{1}{10}$, but he made an error. His work is shown.

$$3\frac{5}{6} \cdot 4\frac{1}{10} = 12\frac{5}{60} = 12\frac{1}{12}$$

1. What error did Mateo make?
2. What is something that Mateo did correctly?
3. Help Mateo understand and learn from his mistake. Write a brief note to Mateo explaining the mistake he made and how to fix it moving forward.

Means

I

Start

To

Acquire

Knowledge

Experience

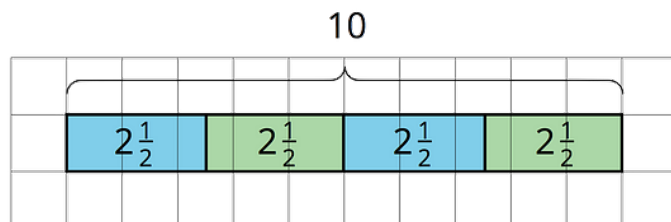
Skills

4. Find the product of $3\frac{5}{6} \cdot 4\frac{1}{10}$. Show your work.



1.7.2 Guided Instruction: Dividing Fractions

Recall how tape diagrams can be used to illustrate division.



- Write the division equation represented by the tape diagram.

dividend **divisor** **quotient**
 _____ ÷ _____ = _____

- How many groups of $2\frac{1}{2}$ are in 10?

- Complete the table by finding each quotient. Use the tape diagram shown.

Tape Diagram	Division Expression	Quotient
	$\frac{1}{2} \div \frac{1}{3}$	
	$3\frac{1}{4} \div \frac{3}{4}$	

- Jacquelyn is making ground beef flautas, where each flauta will use $\frac{3}{10}$ pounds of ground beef. She has a total of $1\frac{4}{5}$ pounds of ground beef to use.

- Which division statement below represents the number of flautas Jacquelyn can make?

$$\frac{3}{10} \div 1\frac{4}{5} \qquad 1\frac{4}{5} \div \frac{3}{10}$$

b. Draw a tape diagram to illustrate the division statement you chose.

c. Solve the division statement.

d. How many $\frac{3}{10}$ pound flautas can she make?

You previously developed an algorithm to use when dividing fractions.



When **dividing fractions**, the division statement can be rewritten as an equivalent multiplication sentence by using the **reciprocal**, or multiplicative inverse, of the second fraction.

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8} = 1\frac{7}{8}$$



Existence of multiplicative inverses

For every $a \neq 0$ there exists $\frac{1}{a}$ so that $a \times \frac{1}{a} = 1$ and $\frac{1}{a} \times a = 1$.

With the tape diagrams, you determined $3\frac{1}{4} \div \frac{3}{4} = 4\frac{1}{3}$.

The standard algorithm can also be used to determine the quotient.

Write the given division expression.	$3\frac{1}{4} \div \frac{3}{4}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{13}{4} \div \frac{3}{4}$
Rewrite the division using multiplication by the reciprocal of the second fraction.	$\frac{13}{4} \times \frac{4}{3}$
Multiply the fractions and simplify.	$\frac{52}{12} = \frac{13}{3} = 4\frac{1}{3}$



5. Find the quotient $5\frac{2}{3} \div 2\frac{4}{9}$ using the algorithm.



1.7.3 Guided Practice: Dividing Fractions

1. Complete the table to find each quotient.

Division Sentence	Rewritten as Fractions (If Needed)	Equivalent Multiplication Sentence	Quotient
$\frac{7}{8} \div 2\frac{3}{10}$			
$\frac{2}{7} \div \frac{3}{4}$			
$8\frac{2}{3} \div \frac{5}{6}$			
$4\frac{4}{5} \div 1\frac{1}{2}$			

2. Find each quotient using your preferred method. Simplify when possible.

a. $\frac{3}{4} \div \frac{9}{14}$

b. $6\frac{7}{8} \div \frac{5}{7}$



1.7.4 Your Turn!

1. Find each quotient. Simplify when possible.

a. $\frac{1}{18} \div \frac{1}{9}$

b. $2\frac{2}{5} \div \frac{1}{10}$

c. $\frac{3}{15} \div 6$

d. $9 \div \frac{3}{2}$

e. $1\frac{5}{9} \div 5\frac{5}{8}$

f. $\frac{1}{20} \div \frac{1}{4}$

g. $\frac{4}{12} \div \frac{1}{7}$

h. $2 \div 3\frac{4}{15}$

i. $1\frac{6}{10} \div 5\frac{3}{4}$



1.7.5 Wrap-Up: Ticket Out the Door

1. Find each product or quotient. Simplify when possible.

a. $\frac{2}{11} \div \frac{1}{3}$

b. $\left(\frac{2}{23}\right)\left(3\frac{3}{23}\right)$

c. $2\frac{27}{32} \times \frac{3}{182}$

d. $3\frac{3}{4} \div 4\frac{1}{6}$